



2 Real Analysis 2023

Tutorial 5

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The monotone convergence theorem

Problem 17 Prove that if a sequence is decreasing and bounded below, then it converges, and its limit is its infimum.

Problem 18 Let $(x_n)_{n=0}^{\infty}$ be a sequence defined recursively by

$$x_0 = 0, \quad x_n = \frac{1}{3}(2 + x_{n-1}^2), \quad n \in \mathbb{N}.$$

- (a) Show that $0 \leq x_n \leq 1$ for all $n = 0, 1, 2, \dots$
- (b) Show that the sequence $(x_n)_{n=0}^{\infty}$ is increasing.
- (c) Justify that the sequence $(x_n)_{n=0}^{\infty}$ converges.
- (d) Find $\lim_{n \rightarrow \infty} x_n$.

Problem 19 We define the sequence $(u_n)_{n=1}^{\infty}$ recursively by

$$u_1 = \sqrt{2}, \quad u_{n+1} = \sqrt{2 + u_n}, \quad n \in \mathbb{N}.$$

- (a) Prove that the sequence $(u_n)_{n=1}^{\infty}$ converges.
- (b) Find $\lim_{n \rightarrow \infty} u_n$.

Series

Problem 20 For each of the series, find an expression for the partial sum. Determine whether the series converges or diverges. If the series converges, calculate its sum.

(a) $\sum_{n=0}^{\infty} \left(-\frac{2}{3}\right)^n,$	(b) $\sum_{n=1}^{\infty} 2^{-n} 5^{n-1},$	(c) $\sum_{n=0}^{\infty} \left(\frac{5}{n+2} - \frac{5}{n+3}\right),$
(d) $\sum_{n=1}^{\infty} \frac{1}{(n+3)(n+4)},$	(e) $\sum_{n=0}^{\infty} \frac{2}{3^{2n+1}},$	(f) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n} + \sqrt{n+1}}.$